

## INCHEON INTL, Korea, Republic of

WMO: 471131

Lat: 37.469N

Lon: 126.451E

Elev: 7

StdP: 101.24

Time Zone: 9.00 (E09)

Period: 04-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -10.9      | -9.0 | -21.0                         | 0.6 | -7.8 | -19.1 | 0.7 | -6.8 | 11.6                  | -6.3 | 10.4 | -5.2 | 4.9                   | 320  | 0.485 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 5.7                    | 31.8            | 24.3 | 30.1 | 24.0 | 29.0 | 23.4 | 26.0                | 28.6 | 25.4 | 28.2 | 24.7 | 27.6 | 4.5                  | 300  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 25.1                            | 20.3 | 27.4 | 24.2 | 19.2 | 26.9 | 23.9 | 18.8 | 26.7 | 80.6          | 28.9 | 77.3 | 27.6 | 74.7 | 28.0 | 27.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 10.7              | 9.4  | 8.3 | -12.7                      | 33.7 | 2.7                | 1.0 | -14.7                                              | 34.5 | -16.3      | 35.1 | -17.8      | 35.7 | -19.8      | 36.4 |    |    |
|                   |      |     | -14.1                      | 26.5 | 2.4                | 0.9 | -15.8                                              | 27.1 | -17.3      | 27.7 | -18.6      | 28.2 | -20.4      | 28.9 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 12.5   | -2.3  | -0.1  | 5.2   | 11.0  | 16.8  | 21.4  | 24.8  | 26.3  | 22.0  | 15.8  | 8.1   | 0.2  |
|                                                                                  | DBStd       | 10.27  | 3.79  | 3.75  | 3.20  | 2.92  | 2.44  | 1.92  | 1.92  | 2.19  | 2.18  | 3.20  | 4.33  | 4.40 |
|                                                                                  | HDD10.0     | 1233   | 381   | 282   | 154   | 22    | 0     | 0     | 0     | 0     | 0     | 3     | 86    | 304  |
|                                                                                  | HDD18.3     | 2808   | 639   | 516   | 409   | 221   | 60    | 2     | 0     | 2     | 89    | 307   | 563   |      |
|                                                                                  | CDD10.0     | 2146   | 0     | 0     | 3     | 52    | 211   | 342   | 459   | 506   | 361   | 182   | 29    | 0    |
|                                                                                  | CDD18.3     | 679    | 0     | 0     | 0     | 1     | 13    | 94    | 201   | 248   | 113   | 11    | 0     | 0    |
|                                                                                  | CDH23.3     | 4383   | 0     | 0     | 0     | 4     | 51    | 348   | 1269  | 2233  | 460   | 18    | 0     | 0    |
|                                                                                  | CDH26.7     | 1021   | 0     | 0     | 0     | 1     | 2     | 36    | 275   | 658   | 48    | 1     | 0     | 0    |
| Wind                                                                             | WSAvg       | 3.8    | 3.8   | 3.9   | 4.3   | 4.0   | 3.7   | 3.3   | 3.6   | 3.6   | 3.4   | 3.7   | 4.1   | 4.2  |
| Precipitation                                                                    | PrecAvg     | 1358   | 17    | 27    | 39    | 77    | 104   | 140   | 398   | 291   | 149   | 50    | 45    | 23   |
|                                                                                  | PrecMax     | 2211   | 50    | 61    | 106   | 164   | 265   | 502   | 958   | 785   | 558   | 214   | 110   | 63   |
|                                                                                  | PrecMin     | 755    | 2     | 1     | 5     | 17    | 14    | 33    | 165   | 41    | 12    | 2     | 11    | 3    |
|                                                                                  | PrecStd     | 327    | 14    | 19    | 25    | 34    | 55    | 101   | 177   | 190   | 133   | 45    | 24    | 15   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 8.2   | 12.9  | 17.0  | 22.2  | 26.8  | 29.1  | 32.9  | 34.2  | 29.2  | 25.2  | 19.2  | 12.0 |
|                                                                                  |             | MCWB   | 5.2   | 8.5   | 10.9  | 13.8  | 16.5  | 19.5  | 23.3  | 23.9  | 22.1  | 17.9  | 14.9  | 8.4  |
|                                                                                  | 2%          | DB     | 6.1   | 9.1   | 14.0  | 19.8  | 24.9  | 27.8  | 30.2  | 32.2  | 28.0  | 23.1  | 17.9  | 9.9  |
|                                                                                  |             | MCWB   | 3.6   | 5.7   | 9.5   | 12.4  | 16.3  | 19.6  | 23.9  | 24.4  | 21.0  | 17.1  | 14.1  | 7.1  |
|                                                                                  | 5%          | DB     | 4.8   | 7.0   | 11.9  | 18.0  | 23.1  | 26.8  | 29.1  | 31.1  | 27.0  | 22.0  | 16.0  | 8.1  |
|                                                                                  |             | MCWB   | 2.2   | 3.9   | 7.9   | 11.2  | 15.3  | 19.4  | 23.3  | 24.7  | 20.5  | 16.5  | 12.6  | 5.5  |
|                                                                                  | 10%         | DB     | 3.2   | 5.2   | 10.1  | 16.1  | 21.8  | 25.2  | 28.1  | 30.0  | 25.9  | 20.8  | 14.8  | 6.8  |
|                                                                                  |             | MCWB   | 0.8   | 2.5   | 6.3   | 10.4  | 14.8  | 19.1  | 23.0  | 24.2  | 19.8  | 15.9  | 11.3  | 4.3  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 5.7   | 9.8   | 12.5  | 15.7  | 19.4  | 23.6  | 26.0  | 27.1  | 24.8  | 20.4  | 16.2  | 9.6  |
|                                                                                  |             | MCDB   | 7.4   | 11.9  | 15.5  | 20.2  | 22.9  | 26.3  | 28.5  | 30.0  | 27.3  | 22.7  | 18.5  | 11.1 |
|                                                                                  | 2%          | WB     | 4.0   | 5.8   | 10.5  | 13.8  | 17.9  | 22.3  | 25.2  | 26.4  | 23.3  | 19.2  | 14.8  | 7.2  |
|                                                                                  |             | MCDB   | 5.6   | 7.8   | 12.9  | 17.8  | 21.4  | 24.8  | 28.2  | 29.1  | 25.3  | 21.7  | 16.9  | 9.3  |
|                                                                                  | 5%          | WB     | 2.5   | 4.3   | 8.5   | 12.5  | 16.9  | 21.2  | 24.6  | 25.9  | 22.2  | 18.0  | 12.9  | 5.7  |
|                                                                                  |             | MCDB   | 4.2   | 6.4   | 10.9  | 15.9  | 20.9  | 24.1  | 27.6  | 28.5  | 25.0  | 21.1  | 15.4  | 7.6  |
|                                                                                  | 10%         | WB     | 1.2   | 3.0   | 6.8   | 11.2  | 16.1  | 20.5  | 24.1  | 25.4  | 21.4  | 16.7  | 11.5  | 4.3  |
|                                                                                  |             | MCDB   | 3.0   | 5.1   | 9.1   | 14.8  | 19.9  | 23.5  | 26.8  | 28.2  | 24.5  | 19.7  | 14.4  | 6.3  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 6.8   | 7.3   | 7.7   | 8.2   | 8.4   | 7.3   | 5.6   | 5.7   | 6.6   | 7.5   | 6.9   | 6.8  |
|                                                                                  |             | MCDBR  | 7.7   | 9.5   | 9.7   | 10.9  | 10.5  | 8.8   | 7.1   | 6.9   | 7.6   | 8.2   | 7.3   | 8.1  |
|                                                                                  |             | MCWBR  | 6.2   | 6.6   | 5.8   | 5.1   | 4.5   | 3.5   | 2.9   | 2.9   | 3.6   | 4.5   | 5.6   | 6.8  |
|                                                                                  | 5% WB       | MCDBR  | 7.3   | 8.7   | 8.6   | 8.9   | 8.6   | 6.6   | 5.8   | 5.6   | 6.2   | 6.6   | 7.0   | 7.6  |
|                                                                                  |             | MCWBR  | 6.4   | 6.7   | 6.2   | 5.5   | 4.7   | 3.5   | 3.0   | 2.9   | 3.7   | 4.8   | 6.2   | 7.0  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.439  | 0.492 | 0.570 | 0.615 | 0.630 | 0.705 | 0.671 | 0.584 | 0.516 | 0.508 | 0.489 | 0.440 |      |
|                                                                                  | taud        | 1.952  | 1.857 | 1.722 | 1.676 | 1.698 | 1.620 | 1.736 | 1.942 | 2.047 | 1.934 | 1.926 | 1.967 |      |
|                                                                                  | Ebn at Noon | 694    | 709   | 694   | 695   | 695   | 643   | 661   | 712   | 737   | 686   | 636   | 658   |      |
|                                                                                  | Edh at Noon | 139    | 173   | 217   | 240   | 239   | 258   | 228   | 182   | 155   | 158   | 142   | 128   |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.52   | 3.29  | 4.37  | 5.07  | 5.68  | 5.31  | 4.21  | 4.51  | 4.25  | 3.65  | 2.45  | 2.17  |      |
|                                                                                  | RadStd      | 0.16   | 0.26  | 0.41  | 0.32  | 0.55  | 0.61  | 0.66  | 0.63  | 0.43  | 0.25  | 0.24  | 0.14  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (39 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +91     | +53     |

Nomenclature: See separate page